Supplementary Material

Confidence Calibration and Uncertainty-Aware Prediction for Reliable Skin Lesion Classification

R. Mahajan, P. Paliwal, Z. Shah, R. K. Nagra

This document extends the main paper with details that did not fit within its page limit. All numbers are generated from saved per-image predictions by the scripts in the code repository. E0/E1: ResNet-50 trained with plain or class-weighted cross-entropy; seed 42 is the model of the submitted manuscript. Source collections: HAM10000 [1], BCN_20000 [2] and MSK [3].

S1 Data, Sources and Duplicate Audit

Lesion identifiers are available for 23,247 of the 25,331 images. They cover all HAM10000 and BCN_20000 images and 819 MSK images; the remaining 2,084 MSK images have no identifier and were treated as single-image groups. The split uses 13,931 groups, 5,059 of which contain more than one image; no group occurs in two partitions.

Duplicate screen. Exact duplicates were found by MD5 hashes of the files and of the decoded 224×224 pixels: 50 groups of pixel-identical images, 4 of which cross partitions. Near-duplicates were found with a 64-bit DCT perceptual hash and the cosine similarity of ImageNet ResNet-50 embeddings. Candidate pairs had a hash distance ≤ 6 or a cosine ≥ 0.95, and a pair was flagged when the hash distance was ≤ 4 and the cosine ≥ 0.97. Within partitions, 97% of the 428 flagged pairs share a lesion identifier, which validates the rule. Across partitions it flags 10 pairs (7 near pixel-identical) involving 8 internal-test images. Visual inspection showed these to be re-uploads of the same image under different identifiers. Removing them gives balanced accuracy 0.565 (all images: 0.5653). Removing every test image with any cross-partition candidate of cosine ≥ 0.97 (276 images) gives 0.576. Between the internal data and the official test set there are no exact duplicates and 2 flagged near-duplicates; excluding them changes the official known-class balanced accuracy from 0.491 to 0.491.

S2 Training Configuration

Table S1: Training configuration (identical for all runs except the seed).

Input	224×224 RGB, bicubic resize, ImageNet mean/std
Augmentation (train)	horizontal flip, rotation $\pm 15^\circ$, scale $\pm 10\%$, translation $\pm 5\%$, brightness/contrast ± 0.1
Optimiser	AdamW, weight decay 10^{-4}, batch 32, gradient clipping 1.0, no label smoothing
Phase 1	backbone frozen, 10 epochs, learning rate 10^{-3}
Phase 2	last 30% of backbone parameters unfrozen, up to 15 epochs, learning rate 5×10^{-5}
Schedule / stopping	ReduceLROnPlateau on validation macro-F1 (factor 0.5, patience 3); early stopping on validation macro-F1 (patience 6)
Class weights (E1)	$w_c = N/(Kn_c)$ from the training split: NV 0.24, MEL 0.71, BCC 0.96, BKL 1.20, AK 3.70, SCC 5.01, VASC 12.2, DF 12.8
Hardware / software	NVIDIA RTX 4050 Laptop GPU (6 GB), PyTorch 2.11, CUDA 12.8; about 1 h per run

Seed 42 corresponds to the models of the submitted manuscript; seeds 43–46 were retrained with an identical protocol. For speed, training read images from a lossless cache of the 224×224 resize. This is bit-identical to resizing on the fly, as verified on 200 images.

S3 Checkpoint Rule and Seed-Level Statistics

The stored weights of the submitted models are those of the epoch at which early stopping triggered: epoch 24 for E1 (best validation macro-F1 at epoch 18) and epoch 21 for E0 (best at epoch 15). EfficientNet-B3 ran all 25 epochs, and its last epoch was also its best. For the new seeds both checkpoints were stored. The two rules differ only when early stopping triggers before the last epoch, which happened in 1 of 4 new E1 runs. Averaged over those runs, the best-epoch weights had a lower temperature (2.308 vs. 2.530) and a lower uncalibrated ECE (0.114 vs. 0.131). Balanced accuracy (0.568 vs. 0.572), AURC (0.132 vs. 0.133) and calibrated NLL (0.816 vs. 0.815) were similar.

Table S2: Mean ± SD over independently trained seeds (seed 42 = submitted run; further seeds retrained with the identical protocol). All post-hoc maps and thresholds fitted on validation. Last column: mean per-seed difference, with (number of seeds where E1 > E0 / seeds paired, paired t-test p). †Official ISIC 2019 test set, never used during development (8 known classes; UNK = out-of-distribution).

Metric (internal test unless marked)	E0 ResNet-50 (n=5)	E1 ResNet-50 (n=5)	E1 EffNet-B3 (n=3)	E1−E0 paired
Accuracy	0.726 ± 0.005	0.719 ± 0.010	0.684 ± 0.005	-0.008 (2/5, $p = 0.15$)
Balanced accuracy	0.532 ± 0.006	0.571 ± 0.013	0.543 ± 0.008	+0.039 (5/5, $p = 0.004$)
Macro-F1	0.543 ± 0.011	0.555 ± 0.012	0.502 ± 0.013	+0.011 (4/5, $p = 0.09$)
ROC-AUC	0.926 ± 0.003	0.922 ± 0.005	0.920 ± 0.001	-0.004 (1/5, $p = 0.08$)
ECE (uncalibrated)	0.188 ± 0.009	0.135 ± 0.019	0.083 ± 0.004	-0.053 (0/5, $p = 0.009$)
ECE (T)	0.026 ± 0.005	0.049 ± 0.011	0.043 ± 0.011	+0.024 (5/5, $p = 0.009$)
NLL (T)	0.762 ± 0.009	0.814 ± 0.005	0.870 ± 0.016	+0.052 (5/5, $p < 0.001$)
Brier (T)	0.380 ± 0.003	0.401 ± 0.006	0.438 ± 0.006	+0.021 (5/5, $p = 0.002$)
Temperature T	3.209 ± 0.247	2.606 ± 0.222	1.660 ± 0.100	-0.603 (0/5, $p = 0.02$)
Error AUROC, MSP	0.792 ± 0.006	0.762 ± 0.006	0.761 ± 0.012	-0.030 (0/5, $p = 0.001$)
Error AUROC, MC entropy	0.778 ± 0.008	0.750 ± 0.007	0.761 ± 0.011	-0.028 (0/5, $p < 0.001$)
AURC, MSP	0.113 ± 0.004	0.133 ± 0.005	0.152 ± 0.006	+0.020 (5/5, $p < 0.001$)
Coverage at val. 85%	0.870 ± 0.010	0.870 ± 0.013	0.869 ± 0.004	+0.000 (2/5, $p = 0.99$)
Selective accuracy	0.776 ± 0.004	0.765 ± 0.009	0.731 ± 0.004	-0.012 (1/5, $p = 0.07$)
Official test bal. acc.†	0.472 ± 0.012	0.508 ± 0.014	0.523 ± 0.002	+0.037 (5/5, $p = 0.02$)
Official test ECE (T)†	0.021 ± 0.004	0.033 ± 0.009	0.019 ± 0.006	+0.012 (4/5, $p = 0.05$)
UNK detection AUROC (MSP)†	0.664 ± 0.019	0.624 ± 0.013	0.618 ± 0.004	-0.041 (0/5, $p = 0.007$)

Table S3: Which model is best depends on the reliability dimension (means over seeds; internal test unless stated).

Dimension	R50-E0	R50-E1	B3-E1	Best
Balanced accuracy	0.532	0.571	0.543	R50-E1
Accuracy	0.726	0.719	0.684	R50-E0
Macro-F1	0.543	0.555	0.502	R50-E1
ROC-AUC	0.926	0.922	0.920	R50-E0
ECE, uncalibrated	0.188	0.135	0.083	B3-E1
ECE after T	0.026	0.049	0.043	R50-E0
NLL after T	0.762	0.814	0.870	R50-E0
Error AUROC (MSP)	0.792	0.762	0.761	R50-E0
AURC (MSP)	0.113	0.133	0.152	R50-E0
Official test bal. acc.	0.472	0.508	0.523	B3-E1
Official test ECE_T	0.021	0.033	0.019	B3-E1
UNK detection AUROC	0.664	0.624	0.618	R50-E0

S4 Calibration Details

ECE uses 15 equal-width bins; adaptive ECE (aECE) uses 15 equal-mass bins and class-wise ECE (cwECE) averages the ECE of each class probability [4, 5]. Prior correction subtracts $\log w_c$ from the logits of E1 (and logit adjustment adds it to E0) following [6]. Calibration-aware training losses such as focal loss [7] were not evaluated.

Table S4: Calibration on the internal test split (seed 42). All maps are fitted on validation only. ECE: 15 equal-width bins; aECE: 15 equal-mass bins; cwECE: class-wise ECE. Vector scaling and prior correction change the decision rule, hence the last column.

Model	Post-hoc map	ECE	aECE	cwECE	Brier	NLL	Bal. acc.
ResNet-50 E1	none	0.149	0.148	0.042	0.425	1.264	0.565
	temperature	0.050	0.047	0.032	0.394	0.811	0.565
	vector scaling	0.038	0.041	0.015	0.375	0.768	0.540
	prior correction	0.139	0.139	0.038	0.409	1.305	0.533
	prior corr. + T	0.065	0.066	0.031	0.384	0.803	0.533
ResNet-50 E0	none	0.179	0.177	0.046	0.438	1.229	0.537
	temperature	0.031	0.029	0.014	0.375	0.753	0.537
	vector scaling	0.025	0.023	0.011	0.375	0.751	0.505
EffNet-B3 E1	none	0.086	0.087	0.041	0.452	0.990	0.536
	temperature	0.055	0.057	0.038	0.443	0.882	0.536
	vector scaling	0.024	0.024	0.014	0.392	0.792	0.444
	prior correction	0.092	0.089	0.026	0.409	0.936	0.457
	prior corr. + T	0.046	0.048	0.026	0.401	0.819	0.457

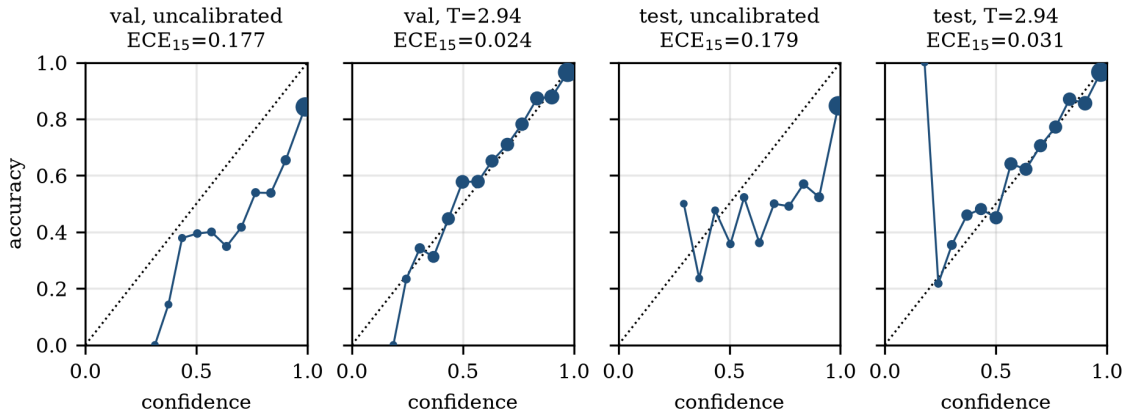


Figure S1: Reliability diagrams of ResNet-50 E0 (seed 42) on validation and internal test, before and after temperature scaling.

S5 Selective Prediction Details

Table S5: Validation-locked selective prediction (ResNet-50 E1, seed 42). Thresholds are chosen on validation for a target validation coverage and applied unchanged to test. Rej. acc.: accuracy among rejected test cases; MEL sens.: melanoma sensitivity among accepted cases (all cases: 0.693).

Score	Target	τ	Val. cov.	Test cov.	Sel. acc.	Rej. acc.	MEL sens.	MEL rej.
MSP (cal.)	0.70	0.524	0.690	0.723	0.812	0.493	0.756	147
MSP (cal.)	0.80	0.463	0.802	0.841	0.786	0.397	0.742	72
MSP (cal.)	0.85	0.429	0.851	0.884	0.769	0.380	0.733	53
MSP (cal.)	0.90	0.391	0.900	0.922	0.752	0.383	0.711	32
MSP (cal.)	0.95	0.348	0.941	0.950	0.744	0.333	0.706	18
MSP (raw)	0.70	0.798	0.691	0.734	0.809	0.487	0.790	161
MSP (raw)	0.80	0.674	0.801	0.836	0.781	0.434	0.755	95
MSP (raw)	0.85	0.616	0.851	0.878	0.764	0.432	0.732	66
MSP (raw)	0.90	0.564	0.901	0.912	0.756	0.395	0.728	47
MSP (raw)	0.95	0.518	0.941	0.947	0.745	0.338	0.709	18
MC entropy	0.70	1.187	0.700	0.746	0.798	0.506	0.741	113
MC entropy	0.80	1.330	0.800	0.828	0.773	0.488	0.726	72
MC entropy	0.85	1.419	0.850	0.874	0.763	0.455	0.719	53
MC entropy	0.90	1.517	0.900	0.922	0.749	0.423	0.704	32
MC entropy	0.95	1.622	0.950	0.956	0.737	0.432	0.697	16

Table S6: Risk-controlled thresholds (SGR, Geifman & El-Yaniv) fitted on validation with $\delta = 0.05$ and applied to test.

Model	r^*	θ	Bound	Val. cov.	Test cov.	Test risk
ResNet-50 E1	0.10	0.770	0.099	0.342	0.386	0.086
ResNet-50 E1	0.15	0.643	0.150	0.503	0.555	0.136
ResNet-50 E1	0.20	0.532	0.200	0.674	0.710	0.181
ResNet-50 E1	0.25	0.457	0.250	0.813	0.848	0.218
ResNet-50 E0	0.10	0.806	0.099	0.421	0.434	0.079
ResNet-50 E0	0.15	0.667	0.150	0.588	0.607	0.130
ResNet-50 E0	0.20	0.548	0.200	0.731	0.753	0.172
ResNet-50 E0	0.25	0.417	0.250	0.886	0.897	0.229
EffNet-B3 E1	0.10	0.791	0.100	0.299	0.327	0.117
EffNet-B3 E1	0.15	0.651	0.150	0.457	0.487	0.151
EffNet-B3 E1	0.20	0.576	0.200	0.560	0.584	0.179
EffNet-B3 E1	0.25	0.496	0.250	0.689	0.711	0.212

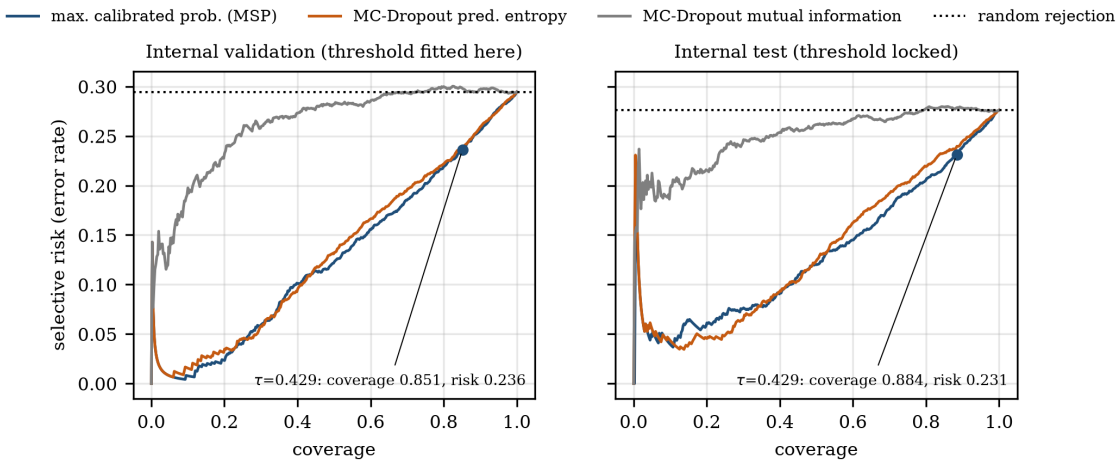


Figure S2: Risk–coverage curves of ResNet-50 E1 (seed 42) on validation, where the threshold is fitted, and on the internal test split, where it is locked.

S6 Uncertainty Scores: Full Table

Mutual information and expected entropy follow the standard decomposition of predictive entropy [8]. Conformal sets (main paper) use the LAC score [9], the adaptive APS score [10] and a class-conditional (Mondrian) variant of LAC. Selective risk is summarised by AURC and E-AURC [11], and risk-controlled thresholds use SGR [12].

Table S7: Error detection on the internal test split (ResNet-50 E1, seed 42). Target: errors of the deployed (deterministic) classifier. Δ: paired lesion-cluster bootstrap difference to calibrated MSP. pAUROC: standardised partial AUROC for FPR ≤ 0.2. AURC: area under the risk–coverage curve (random rejection = 0.276).

Score	AUROC	Δ vs MSP [95% CI]	pAUROC	AURC
MSP ($1 - \max_k p_k$), calibrated	0.761	–	0.627	0.134
MSP, uncalibrated	0.745	−0.016 [−0.025, −0.007]	0.621	0.144
Softmax entropy, calibrated	0.750	−0.010 [−0.022, +0.001]	0.601	0.134
MC predictive entropy (paper)	0.745	−0.016 [−0.029, −0.003]	0.598	0.136
MC expected entropy (aleatoric)	0.747	−0.014 [−0.027, −0.002]	0.600	0.136
MC mutual information (epistemic)	0.528	−0.232 [−0.276, −0.189]	0.497	0.249
MC predictive variance	0.620	−0.141 [−0.178, −0.107]	0.534	0.197
MC mean-probability MSP	0.761	+0.000 [−0.003, +0.003]	0.625	0.134
TTA predictive entropy	0.760	−0.000 [−0.016, +0.016]	0.615	0.130
TTA mutual information	0.705	−0.056 [−0.085, −0.028]	0.580	0.154

S7 Grad-CAM Checks

The submitted version showed four curated Grad-CAM examples [13]. We applied two quantitative checks to 300 random test images, at the original layer (`conv3` of the last bottleneck) and at the output of `layer4`.

- *Cascading parameter randomisation* [14]. Re-initialising only the head already drops the rank correlation between original and randomised maps to 0.04, so the maps depend on the trained weights (Table S8).
- *Deletion/insertion* [15]. Grad-CAM clearly beats a random map. For deletion, however, it is no better than a fixed centre-Gaussian prior that ignores the image (AUC 0.538 vs. 0.538, Wilcoxon $p = 0.861$); for insertion it is only modestly better (0.754 vs. 0.723, $p = 0.002$) (Table S9).

On average, 55.5% of the map mass falls in the central quarter of the image. Maps of uncertain predictions are more concentrated (Spearman $\rho = -0.58$ between map entropy and MC entropy) and often lie off the lesion (Fig. S3). As an error detector, map concentration reaches an AUROC of only 0.627. Grad-CAM is therefore not used as reliability evidence in the main paper.

Table S8: Cascading model-parameter randomisation (300 random test images): similarity between Grad-CAM maps of the trained and the progressively randomised network. Values near 0 indicate model-sensitive maps.

Randomised	Spearman ρ	Top-10% IoU
up to head	0.043	0.166
up to layer4	0.115	0.078
up to layer3	−0.034	0.048
up to layer2	0.022	0.053
up to layer1	−0.094	0.041

Table S9: Deletion/insertion faithfulness (300 random test images; probability of the predicted class as salient pixels are blurred out / restored in 5% steps). Lower deletion and higher insertion AUC are better.

Saliency	Deletion AUC	Insertion AUC
Grad-CAM (paper layer)	0.538	0.754
Grad-CAM (layer4 output)	0.533	0.760
Dataset-mean Grad-CAM	0.540	0.731
Centre Gaussian prior	0.538	0.723
Random map	0.586	0.579

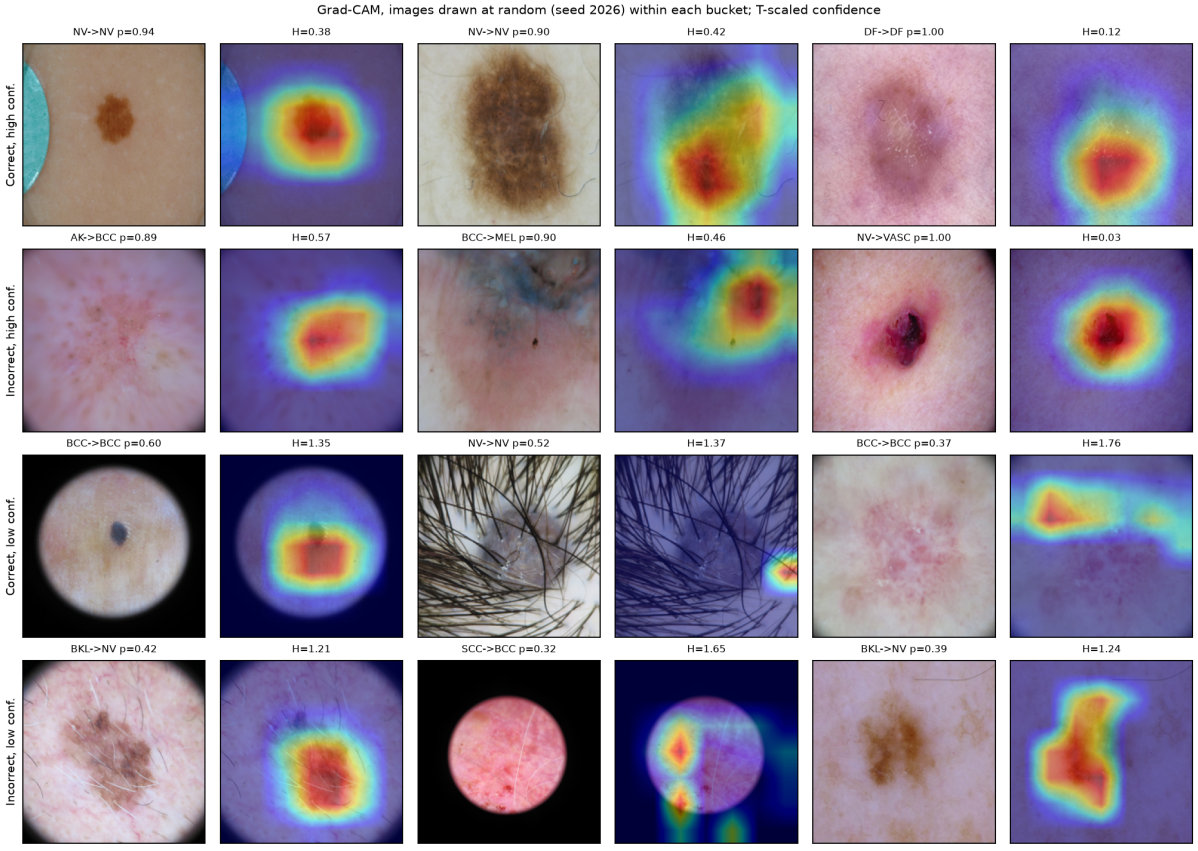


Figure S3: Grad-CAM maps for test images drawn at random (fixed seed) from each correctness/confidence bucket of ResNet-50 E1. Titles: true→predicted class, calibrated confidence and MC entropy H.

S8 Higher-Resolution Replication

To test whether the conclusions depend on the 224×224 input, ResNet-50 E0 and E1 were retrained at 320×320 with otherwise identical settings (Table S10).

Table S10: ResNet-50 at 224×224 vs. 320×320 px (mean±SD over seeds; internal test unless stated). Same protocol; only the input resolution changes.

Metric	E0 224 (n=5)	E1 224 (n=5)	E0 320 (n=2)	E1 320 (n=2)
Balanced accuracy	0.532±0.006	0.571±0.013	0.572±0.017	0.596±0.032
Accuracy	0.726±0.005	0.719±0.010	0.749±0.003	0.740±0.015
ECE, uncalibrated	0.188±0.009	0.135±0.019	0.152±0.001	0.129±0.020
ECE, after T	0.026±0.005	0.049±0.011	0.052±0.005	0.057±0.001
NLL, after T	0.762±0.009	0.814±0.005	0.727±0.009	0.774±0.027
Temperature	3.209±0.247	2.606±0.222	3.094±0.245	2.733±0.465
Error AUROC (MSP)	0.792±0.006	0.762±0.006	0.776±0.004	0.761±0.011
Error AUROC (MC entropy)	0.778±0.008	0.750±0.007	0.753±0.009	0.753±0.017
AURC (MSP)	0.113±0.004	0.133±0.005	0.107±0.004	0.119±0.011
Official bal. acc.	0.472±0.012	0.508±0.014	0.515±0.011	0.530±0.029
Official UNK AUROC (MSP)	0.664±0.019	0.624±0.013	0.696±0.028	0.626±0.012
Official UNK AUROC (MC MI)	0.699±0.017	0.666±0.010	0.699±0.004	0.680±0.004

References

[1] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Scientific Data*, vol. 5, p. 180161, 2018.

[2] M. Combalia, N. C. F. Codella, V. Rotemberg, B. Helba, V. Vilaplana, O. Reiter, C. Carrera, A. Barreiro, A. C. Halpern, S. Puig, and J. Malvehy, "BCN20000: Dermoscopic lesions in the wild," *arXiv preprint arXiv:1908.02288*, 2019.

[3] N. C. F. Codella, D. Gutman, M. E. Celebi, B. Helba, M. A. Marchetti, S. W. Dusza, A. Kalloo, K. Liopyris, N. Mishra, H. Kittler, and A. Halpern, "Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (ISBI), hosted by the international skin imaging collaboration (ISIC)," in *Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI)*, 2018, pp. 168–172.

[4] J. Nixon, M. W. Dusenberry, L. Zhang, G. Jerfel, and D. Tran, "Measuring calibration in deep learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2019.

[5] M. Kull, M. Perello-Nieto, M. Kängsepp, T. Silva Filho, H. Song, and P. Flach, "Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with Dirichlet calibration," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[6] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit, and S. Kumar, "Long-tail learning via logit adjustment," in *International Conference on Learning Representations (ICLR)*, 2021.

[7] J. Mukhoti, V. Kulharia, A. Sanyal, S. Golodetz, P. H. S. Torr, and P. K. Dokania, "Calibrating deep neural networks using focal loss," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[8] S. Depeweg, J. M. Hernández-Lobato, F. Doshi-Velez, and S. Udluft, "Decomposition of uncertainty in Bayesian deep learning for efficient and risk-sensitive learning," in *Proceedings of the 35th International Conference on Machine Learning*, vol. 80, 2018, pp. 1184–1193.

[9] M. Sadinle, J. Lei, and L. Wasserman, "Least ambiguous set-valued classifiers with bounded error levels," *Journal of the American Statistical Association*, vol. 114, no. 525, pp. 223–234, 2019.

[10] Y. Romano, M. Sesia, and E. Candès, "Classification with valid and adaptive coverage," in *Advances in Neural Information Processing Systems*, vol. 33, 2020.

[11] Y. Geifman, G. Uziel, and R. El-Yaniv, "Bias-reduced uncertainty estimation for deep neural classifiers," in *International Conference on Learning Representations (ICLR)*, 2019.

[12] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4878–4887.

[13] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 618–626.

[14] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," in *Advances in Neural Information Processing Systems*, vol. 31, 2018.

[15] V. Petsiuk, A. Das, and K. Saenko, "RISE: Randomized input sampling for explanation of black-box models," in *Proceedings of the British Machine Vision Conference (BMVC)*, 2018.